Chapter 18

Bayesian Statistics

18.1 Bayesian Concepts

The classical methods of estimation that we have studied in this text are based solely on information provided by the random sample. These methods essentially interpret probabilities as relative frequencies. For example, in arriving at a 95% confidence interval for μ, we interpret the statement

$$P(-1.96 < Z < 1.96) = 0.95$$

to mean that 95% of the time in repeated experiments Z will fall between -1.96 and 1.96. Since

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

for a normal sample with known variance, the probability statement here means that 95% of the random intervals $(\bar{X} - 1.96\sigma/\sqrt{n}, \bar{X} + 1.96\sigma/\sqrt{n})$ contain the true mean μ. Another approach to statistical methods of estimation is called **Bayesian methodology**. The main idea of the method comes from Bayes' rule, described in Section 2.7. The key difference between the Bayesian approach and the classical or frequentist approach is that in Bayesian concepts, the parameters are viewed as random variables.

Subjective Probability

Subjective probability is the foundation of Bayesian concepts. In Chapter 2, we discussed two possible approaches to probability, namely the relative frequency and the indifference approaches. The first one determines a probability as a consequence of repeated experiments. For instance, to decide the free-throw percentage of a basketball player, we can record the number of shots made and the total number of attempts this player has made. The probability of hitting a free-throw for this player can be calculated as the ratio of these two numbers. On the other hand, if we have no knowledge of any bias in a die, the probability that a 3 will appear in the next throw will be 1/6. Such an approach to probability interpretation is based on the indifference rule.

However, in many situations, the preceding probability interpretations cannot be applied. For instance, consider the questions "What is the probability that it will rain tomorrow?" "How likely is it that this stock will go up by the end of the month?" and "What is the likelihood that two companies will be merged together?" They can hardly be interpreted by the aforementioned approaches, and the answers to these questions may be different for different people. Yet these questions are constantly asked in daily life, and the approach used to explain these probabilities is called *subjective probability*, which reflects one's subjective opinion.

Conditional Perspective

Recall that in Chapters 9 through 17, all statistical inferences were based on the fact that the parameters are unknown but fixed quantities, apart from those in Section 9.14, in which the parameters were treated as variables and the maximum likelihood estimates (MLEs) were calculated conditioning on the observed sample data. In Bayesian statistics, not only are the parameters treated as variables as in MLE calculation, but also they are treated as random.

Because the observed data are the only experimental results for the practitioner, statistical inference is based on the actual observed data from a given experiment. Such a view is called a *conditional perspective*. Furthermore, in Bayesian concepts, since the parameters are treated as random, a probability distribution can be specified, generally by using the *subjective probability* for the parameter. Such a distribution is called a *prior distribution* and it usually reflects the experimenter's prior belief about the parameter. In the Bayesian perspective, once an experiment is conducted and data are observed, all knowledge about the parameter is contained in the actual observed data and in the prior information.

Bayesian Applications

Although Bayes' rule is credited to Thomas Bayes, Bayesian applications were first introduced by French scientist Pierre Simon Laplace, who published a paper on using Bayesian inference on the unknown binomial proportions (for binomial distribution, see Section 5.2).

Since the introduction of the Markov chain Monte Carlo (MCMC) computational tools for Bayesian analysis in the early 1990s, Bayesian statistics has become more and more popular in statistical modeling and data analysis. Meanwhile, methodology developments using Bayesian concepts have progressed dramatically, and they are applied in fields such as bioinformatics, biology, business, engineering, environmental and ecology science, life science and health, medicine, and many others.

18.2 Bayesian Inferences

Consider the problem of finding a point estimate of the parameter θ for the population with distribution $f(x|\ \theta)$, given θ. Denote by $\pi(\theta)$ the prior distribution of θ. Suppose that a random sample of size n, denoted by $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, is observed.

Definition 18.1: The distribution of θ, given $\mathbf{x}$, which is called the posterior distribution, is given by

$$\pi(\theta|\mathbf{x}) = \frac{f(\mathbf{x}|\theta)\pi(\theta)}{g(\mathbf{x})},$$

where $g(\mathbf{x})$ is the marginal distribution of $\mathbf{x}$.

The marginal distribution of $\mathbf{x}$ in the above definition can be calculated using the following formula:

$$g(\mathbf{x}) = \begin{cases} \sum_{\theta} f(\mathbf{x}|\theta)\pi(\theta), & \theta \text{ is discrete,} \\ \int_{-\infty}^{\infty} f(\mathbf{x}|\theta)\pi(\theta)\ d\theta, & \theta \text{ is continuous.} \end{cases}$$

Example 18.1: Assume that the prior distribution for the proportion of defectives produced by a machine is

p	0.1	0.2
$\pi(p)$	0.6	0.4

Denote by x the number of defectives among a random sample of size 2. Find the posterior probability distribution of p, given that x is observed.

***Solution*:** The random variable X follows a binomial distribution

$$f(x|p) = b(x; 2, p) = \binom{2}{x} p^x q^{2-x}, \quad x = 0, 1, 2.$$

The marginal distribution of x can be calculated as

$$\begin{aligned} g(x) &= f(x|0.1)\pi(0.1) + f(x|0.2)\pi(0.2) \\ &= \binom{2}{x}[(0.1)^x(0.9)^{2-x}(0.6) + (0.2)^x(0.8)^{2-x}(0.4)]. \end{aligned}$$

Hence, for $x = 0, 1, 2$, we obtain the marginal probabilities as

x	0	1	2
$g(x)$	0.742	0.236	0.022

The posterior probability of $p = 0.1$, given x, is

$$\pi(0.1|x) = \frac{f(x|0.1)\pi(0.1)}{g(x)} = \frac{(0.1)^x(0.9)^{2-x}(0.6)}{(0.1)^x(0.9)^{2-x}(0.6) + (0.2)^x(0.8)^{2-x}(0.4)},$$

and $\pi(0.2|x) = 1 - \pi(0.1|x)$.

Suppose that $x = 0$ is observed.

$$\pi(0.1|0) = \frac{f(0 \mid 0.1)\pi(0.1)}{g(0)} = \frac{(0.1)^0(0.9)^{2-0}(0.6)}{0.742} = 0.6550,$$

and $\pi(0.2|0) = 0.3450$. If $x = 1$ is observed, $\pi(0.1|1) = 0.4576$, and $\pi(0.2|1) = 0.5424$. Finally, $\pi(0.1|2) = 0.2727$, and $\pi(0.2|2) = 0.7273$.

The prior distribution for Example 18.1 is discrete, although the natural range of p is from 0 to 1. Consider the following example, where we have a prior distribution covering the whole space for p.

Example 18.2: Suppose that the prior distribution of p is uniform (i.e., $\pi(p) = 1$, for $0 < p < 1$). Use the same random variable X as in Example 18.1 to find the posterior distribution of p.

***Solution*:** As in Example 18.1, we have

$$f(x|p) = b(x; 2, p) = \binom{2}{x} p^x q^{2-x}, \quad x = 0, 1, 2.$$

The marginal distribution of x can be calculated as

$$g(x) = \int_0^1 f(x|p)\pi(p)\ dp = \binom{2}{x} \int_0^1 p^x (1-p)^{2-x}\ dp.$$

The integral above can be evaluated at each x directly as $g(0) = 1/3$, $g(1) = 1/3$, and $g(2) = 1/3$. Therefore, the posterior distribution of p, given x, is

$$\pi(p|x) = \frac{\binom{2}{x} p^x (1-p)^{2-x}}{1/3} = 3\binom{2}{x} p^x (1-p)^{2-x}, \quad 0 < p < 1.$$

The posterior distribution above is actually a beta distribution (see Section 6.8) with parameters $\alpha = x + 1$ and $\beta = 3 - x$. So, if $x = 0$ is observed, the posterior distribution of p is a beta distribution with parameters $(1, 3)$. The posterior mean is $\mu = \frac{1}{1+3} = \frac{1}{4}$ and the posterior variance is $\sigma^2 = \frac{(1)(3)}{(1+3)^2(1+3+1)} = \frac{3}{80}$. ■

Using the posterior distribution, we can estimate the parameter(s) in a population in a straightforward fashion. In computing posterior distributions, it is very helpful if one is familiar with the distributions in Chapters 5 and 6. Note that in Definition 18.1, the *variable* in the posterior distribution is θ, while $\mathbf{x}$ is given. Thus, we can treat $g(\mathbf{x})$ as a constant as we calculate the posterior distribution of θ. Then the posterior distribution can be expressed as

$$\pi(\theta|\mathbf{x}) \propto f(\mathbf{x}|\theta)\pi(\theta),$$

where the symbol "$\propto$" stands for *is proportional to*. In the calculation of the posterior distribution above, we can leave the factors that do not depend on θ out of the normalization constant, i.e., the marginal density $g(\mathbf{x})$.

Example 18.3: Suppose that random variables $X_1, \ldots, X_n$ are independent and from a Poisson distribution with mean λ. Assume that the prior distribution of λ is exponential with mean 1. Find the posterior distribution of λ when $\bar{x} = 3$ with $n = 10$.

***Solution*:** The density function of $\mathbf{X} = (X_1, \ldots, X_n)$ is

$$f(\mathbf{x}|\lambda) = \prod_{i=1}^n e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} = e^{-n\lambda} \frac{\lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!},$$

and the prior distribution is

$$\pi(\theta) = e^{-\lambda}, \text{ for } \lambda > 0.$$

Hence, using Definition 18.1 we obtain the posterior distribution of λ as

$$\pi(\lambda|\mathbf{x}) \propto f(\mathbf{x}|\lambda)\pi(\lambda) = e^{-n\lambda}\frac{\lambda^{\sum_{i=1}^{n} x_i}}{\prod_{i=1}^{n} x_i!}e^{-\lambda} \propto e^{-(n+1)\lambda}\lambda^{\sum_{i=1}^{n} x_i}.$$

Referring to the gamma distribution in Section 6.6, we conclude that the posterior distribution of λ follows a gamma distribution with parameters $1+\sum_{i=1}^{n} x_i$ and $\frac{1}{n+1}$. Hence, we have the posterior mean and variance of λ as $\frac{\sum_{i=1}^{n} x_i+1}{n+1}$ and $\frac{\sum_{i=1}^{n} x_i+1}{(n+1)^2}$. So, when $\bar{x} = 3$ with $n = 10$, we have $\sum_{i=1}^{10} x_i = 30$. Hence, the posterior distribution of λ is a gamma distribution with parameters 31 and 1/11. ∎

From Example 18.3 we observe that sometimes it is quite convenient to use the "proportional to" technique in calculating the posterior distribution, especially when the result can be formed to a commonly used distribution as described in Chapters 5 and 6.

Point Estimation Using the Posterior Distribution

Once the posterior distribution is derived, we can easily use the summary of the posterior distribution to make inferences on the population parameters. For instance, the posterior mean, median, and mode can all be used to estimate the parameter.

Example 18.4: Suppose that $x = 1$ is observed for Example 18.2. Find the posterior mean and the posterior mode.

***Solution*:** When $x = 1$, the posterior distribution of p can be expressed as

$$\pi(p|1) = 6p(1-p), \qquad \text{for} \quad 0 < p < 1.$$

To calculate the mean of this distribution, we need to find

$$\int_0^1 6p^2(1-p)\ dp = 6\left(\frac{1}{3} - \frac{1}{4}\right) = \frac{1}{2}.$$

To find the posterior mode, we need to obtain the value of p such that the posterior distribution is maximized. Taking derivative of $\pi(p)$ with respect to p, we obtain $6 - 12p$. Solving for p in $6 - 12p = 0$, we obtain $p = 1/2$. The second derivative is -12, which implies that the posterior mode is achieved at $p = 1/2$. ∎

Bayesian methods of estimation concerning the mean μ of a normal population are based on the following example.

Example 18.5: If $\bar{x}$ is the mean of a random sample of size n from a normal population with known variance σ^2, and the prior distribution of the population mean is a normal distribution with known mean μ_0 and known variance σ_0^2, then show that the posterior distribution of the population mean is also a normal distribution with

mean μ^* and standard deviation σ^*, where

$$\mu^* = \frac{\sigma_0^2}{\sigma_0^2 + \sigma^2/n}\bar{x} + \frac{\sigma^2/n}{\sigma_0^2 + \sigma^2/n}\mu_0 \qquad \text{and} \qquad \sigma^* = \sqrt{\frac{\sigma_0^2\sigma^2}{n\sigma_0^2 + \sigma^2}}.$$

***Solution*:** The density function of our sample is

$$f(x_1, x_2, \ldots, x_n \mid \mu) = \frac{1}{(2\pi)^{n/2}\sigma^n} \exp\left[-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i - \mu}{\sigma}\right)^2\right],$$

for $-\infty < x_i < \infty$ and $i = 1, 2, \ldots, n$, and the prior is

$$\pi(\mu) = \frac{1}{\sqrt{2\pi}\sigma_0} \exp\left[-\frac{1}{2}\left(\frac{\mu - \mu_0}{\sigma_0}\right)^2\right], \qquad -\infty < \mu < \infty.$$

Then the posterior distribution of μ is

$$\begin{aligned}\pi(\mu|\mathbf{x}) &\propto \exp\left\{-\frac{1}{2}\left[\sum_{i=1}^{n}\left(\frac{x_i - \mu}{\sigma}\right)^2 + \left(\frac{\mu - \mu_0}{\sigma_0}\right)^2\right]\right\} \\ &\propto \exp\left\{-\frac{1}{2}\left[\frac{n(\bar{x} - \mu)^2}{\sigma^2} + \frac{(\mu - \mu_0)^2}{\sigma_0^2}\right]\right\},\end{aligned}$$

due to

$$\sum_{i=1}^{n}(x_i - \mu)^2 = \sum_{i=1}^{n}(x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$$

from Section 8.5. Completing the squares for μ yields the posterior distribution

$$\pi(\mu|\mathbf{x}) \propto \exp\left[-\frac{1}{2}\left(\frac{\mu - \mu^*}{\sigma^*}\right)^2\right],$$

where

$$\mu^* = \frac{n\bar{x}\sigma_0^2 + \mu_0\sigma^2}{n\sigma_0^2 + \sigma^2}, \qquad \sigma^* = \sqrt{\frac{\sigma_0^2\sigma^2}{n\sigma_0^2 + \sigma^2}}.$$

This is a normal distribution with mean μ^* and standard deviation σ^*.

The Central Limit Theorem allows us to use Example 18.5 also when we select sufficiently large random samples ($n \geq 30$ for many engineering experimental cases) from nonnormal populations (the distribution is not very far from symmetric), and when the prior distribution of the mean is approximately normal.

Several comments need to be made about Example 18.5. The posterior mean μ^* can also be written as

$$\mu^* = \frac{\sigma_0^2}{\sigma_0^2 + \sigma^2/n}\bar{x} + \frac{\sigma^2/n}{\sigma_0^2 + \sigma^2/n}\mu_0,$$

which is a weighted average of the sample mean $\bar{x}$ and the prior mean μ_0. Since both coefficients are between 0 and 1 and they sum to 1, the posterior mean μ^* is always

between $\bar{x}$ and μ_0. This means that the posterior estimation of μ is influenced by both $\bar{x}$ and μ_0. Furthermore, the weight of $\bar{x}$ depends on the prior variance as well as the variance of the sample mean. For a large sample problem ($n \to \infty$), the posterior mean $\mu^* \to \bar{x}$. This means that the prior mean does not play any role in estimating the population mean μ using the posterior distribution. This is very reasonable since it indicates that when the amount of data is substantial, information from the data will dominate the information on μ provided by the prior. On the other hand, when the prior variance is large ($\sigma_0^2 \to \infty$), the posterior mean μ^* also goes to $\bar{x}$. Note that for a normal distribution, the larger the variance, the flatter the density function. The flatness of the normal distribution in this case means that there is almost no subjective prior information available on the parameter μ before the data are collected. Thus, it is reasonable that the posterior estimation μ^* only depends on the data value $\bar{x}$.

Now consider the posterior standard deviation σ^*. This value can also be written as

$$\sigma^* = \sqrt{\frac{\sigma_0^2 \sigma^2/n}{\sigma_0^2 + \sigma^2/n}}.$$

It is obvious that the value σ^* is smaller than both σ_0 and $\sigma/\sqrt{n}$, the prior standard deviation and the standard deviation of $\bar{x}$, respectively. This suggests that the posterior estimation is more accurate than both the prior and the sample data. Hence, incorporating both the data and prior information results in better posterior information than using any of the data or prior alone. This is a common phenomenon in Bayesian inference. Furthermore, to compute μ^* and σ^* by the formulas in Example 18.5, we have assumed that σ^2 is known. Since this is generally not the case, we shall replace σ^2 by the sample variance s^2 whenever $n \geq 30$.

Bayesian Interval Estimation

Similar to the classical confidence interval, in Bayesian analysis we can calculate a $100(1-\alpha)\%$ Bayesian interval using the posterior distribution.

Definition 18.2: The interval $a < \theta < b$ will be called a $100(1-\alpha)\%$ **Bayesian interval** for θ if

$$\int_{-\infty}^{a} \pi(\theta|x)\ d\theta = \int_{b}^{\infty} \pi(\theta|x)\ d\theta = \frac{\alpha}{2}.$$

Recall that under the frequentist approach, the probability of a confidence interval, say 95%, is interpreted as a coverage probability, which means that if an experiment is repeated again and again (with considerable unobserved data), the probability that the intervals calculated according to the rule will cover the true parameter is 95%. However, in Bayesian interval interpretation, say for a 95% interval, we can state that the probability of the unknown parameter falling into the calculated interval (which only depends on the observed data) is 95%.

Example 18.6: Supposing that $X \sim b(x; n, p)$, with known $n = 2$, and the prior distribution of p is uniform $\pi(p) = 1$, for $0 < p < 1$, find a 95% Bayesian interval for p.

***Solution*:** As in Example 18.2, when $x = 0$, the posterior distribution is a beta distribution with parameters 1 and 3, i.e., $\pi(p|0) = 3(1-p)^2$, for $0 < p < 1$. Thus, we need to solve for a and b using Definition 18.2, which yields the following:

$$0.025 = \int_0^a 3(1-p)^2 \, dp = 1 - (1-a)^3$$

and

$$0.025 = \int_b^1 3(1-p)^2 \, dp = (1-b)^3.$$

The solutions to the above equations result in $a = 0.0084$ and $b = 0.7076$. Therefore, the probability that p falls into $(0.0084, 0.7076)$ is 95%.

For the normal population and normal prior case described in Example 18.5, the posterior mean μ^* is the Bayes estimate of the population mean μ, and a $100(1-\alpha)\%$ **Bayesian interval** for μ can be constructed by computing the interval

$$\mu^* - z_{\alpha/2}\sigma^* < \mu < \mu^* + z_{\alpha/2}\sigma^*,$$

which is centered at the posterior mean and contains $100(1-\alpha)\%$ of the posterior probability.

Example 18.7: An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed with a standard deviation of 100 hours. Prior experience leads us to believe that μ is a value of a normal random variable with a mean $\mu_0 = 800$ hours and a standard deviation $\sigma_0 = 10$ hours. If a random sample of 25 bulbs has an average life of 780 hours, find a 95% Bayesian interval for μ.

***Solution*:** According to Example 18.5, the posterior distribution of the mean is also a normal distribution with mean

$$\mu^* = \frac{(25)(780)(10)^2 + (800)(100)^2}{(25)(10)^2 + (100)^2} = 796$$

and standard deviation

$$\sigma^* = \sqrt{\frac{(10)^2(100)^2}{(25)(10)^2 + (100)^2}} = \sqrt{80}.$$

The 95% Bayesian interval for μ is then given by

$$796 - 1.96\sqrt{80} < \mu < 796 + 1.96\sqrt{80},$$

or

$$778.5 < \mu < 813.5.$$

Hence, we are 95% sure that μ will be between 778.5 and 813.5.

On the other hand, ignoring the prior information about μ, we could proceed as in Section 9.4 and construct the classical 95% confidence interval

$$780 - (1.96)\left(\frac{100}{\sqrt{25}}\right) < \mu < 780 + (1.96)\left(\frac{100}{\sqrt{25}}\right),$$

or $740.8 < \mu < 819.2$, which is seen to be wider than the corresponding Bayesian interval.

18.3 Bayes Estimates Using Decision Theory Framework

Using Bayesian methodology, the posterior distribution of a parameter can be obtained. Bayes estimates can also be derived using the posterior distribution and a loss function when a loss is incurred. A loss function is a function that describes the cost of a decision associated with an event of interest. Here we only list a few commonly used loss functions and their associated Bayes estimates.

Squared-Error Loss

Definition 18.3: The **squared-error loss function** is

$$L(\theta, a) = (\theta - a)^2,$$

where θ is the parameter (or state of nature) and a an action (or estimate).

A Bayes estimate minimizes the posterior expected loss, given on the observed sample data.

Theorem 18.1: The mean of the posterior distribution $\pi(\theta|x)$, denoted by θ^*, is the **Bayes estimate of $\boldsymbol{\theta}$** under the squared-error loss function.

Example 18.8: Find the Bayes estimates of p, for all the values of x, for Example 18.1 when the squared-error loss function is used.

***Solution*:** When $x = 0$, $p^* = (0.1)(0.6550) + (0.2)(0.3450) = 0.1345$.
When $x = 1$, $p^* = (0.1)(0.4576) + (0.2)(0.5424) = 0.1542$.
When $x = 2$, $p^* = (0.1)(0.2727) + (0.2)(0.7273) = 0.1727$.

Note that the classical estimate of p is $\hat{p} = x/n = 0$, $1/2$, and 1, respectively, for the x values at 0, 1, and 2. These classical estimates are very different from the corresponding Bayes estimates. ∎

Example 18.9: Repeat Example 18.8 in the situation of Example 18.2.

***Solution*:** Since the posterior distribution of p is a $B(x+1, 3-x)$ distribution (see Section 6.8 on page 201), the Bayes estimate of p is

$$p^* = E^{\pi(p|x)}(p) = 3\binom{2}{x}\int_0^1 p^{x+1}(1-p)^{2-x}\,dp,$$

which yields $p^* = 1/4$ for $x = 0$, $p^* = 1/2$ for $x = 1$, and $p^* = 3/4$ for $x = 2$, respectively. Notice that when $x = 1$ is observed, the Bayes estimate and the classical estimate $\hat{p}$ are equivalent. ∎

For the normal situation as described in Example 18.5, the Bayes estimate of μ under the squared-error loss will be the posterior mean μ^*.

Example 18.10: Suppose that the sampling distribution of a random variable, X, is Poisson with parameter λ. Assume that the prior distribution of λ follows a gamma distribution

with parameters (α, β). Find the Bayes estimate of λ under the squared-error loss function.

***Solution*:** Using Example 18.3, we conclude that the posterior distribution of λ follows a gamma distribution with parameters $(x+\alpha, (1+1/\beta)^{-1})$. Using Theorem 6.4, we obtain the posterior mean

$$\hat{\lambda} = \frac{x+\alpha}{1+1/\beta}.$$

Since the posterior mean is the Bayes estimate under the squared-error loss, $\hat{\lambda}$ is our Bayes estimate. ■

Absolute-Error Loss

The squared-error loss described above is similar to the least-squares concept we discussed in connection with regression in Chapters 11 and 12. In this section, we introduce another loss function as follows.

Definition 18.4: The **absolute-error loss function** is defined as

$$L(\theta, a) = |\theta - a|,$$

where θ is the parameter and a an action.

Theorem 18.2: The median of the posterior distribution $\pi(\theta|x)$, denoted by θ^*, is the **Bayes estimate of** θ under the absolute-error loss function.

Example 18.11: Under the absolute-error loss, find the Bayes estimator for Example 18.9 when $x = 1$ is observed.

***Solution*:** Again, the posterior distribution of p is a $B(x+1, 3-x)$. When $x = 1$, it is a beta distribution with density $\pi(p \mid x = 1) = 6x(1-x)$ for $0 < x < 1$ and 0 otherwise. The median of this distribution is the value of p^* such that

$$\frac{1}{2} = \int_0^{p^*} 6p(1-p)\, dp = 3p^{*2} - 2p^{*3},$$

which yields the answer $p^* = \frac{1}{2}$. Hence, the Bayes estimate in this case is 0.5. ■

Exercises

18.1 Estimate the proportion of defectives being produced by the machine in Example 18.1 if the random sample of size 2 yields 2 defectives.

18.2 Let us assume that the prior distribution for the proportion p of drinks from a vending machine that overflow is

p	0.05	0.10	0.15
$\pi(p)$	0.3	0.5	0.2

If 2 of the next 9 drinks from this machine overflow, find

(a) the posterior distribution for the proportion p;

(b) the Bayes estimate of p.

18.3 Repeat Exercise 18.2 when 1 of the next 4 drinks overflows and the uniform prior distribution is

$$\pi(p) = 10, \quad 0.05 < p < 0.15.$$

18.4 Service calls come to a maintenance center according to a Poisson process with λ calls per minute. A data set of 20 one-minute periods yields an average of 1.8 calls. If the prior for λ follows an exponential distribution with mean 2, determine the posterior distribution of λ.

18.5 A previous study indicates that the percentage of chain smokers, p, who have lung cancer follows a beta distribution (see Section 6.8) with mean 70% and standard deviation 10%. Suppose a new data set collected shows that 81 out of 120 chain smokers have lung cancer.

(a) Determine the posterior distribution of the percentage of chain smokers who have lung cancer by combining the new data and the prior information.

(b) What is the posterior probability that p is larger than 50%?

18.6 The developer of a new condominium complex claims that 3 out of 5 buyers will prefer a two-bedroom unit, while his banker claims that it would be more correct to say that 7 out of 10 buyers will prefer a two-bedroom unit. In previous predictions of this type, the banker has been twice as reliable as the developer. If 12 of the next 15 condominiums sold in this complex are two-bedroom units, find

(a) the posterior probabilities associated with the claims of the developer and banker;

(b) a point estimate of the proportion of buyers who prefer a two-bedroom unit.

18.7 The burn time for the first stage of a rocket is a normal random variable with a standard deviation of 0.8 minute. Assume a normal prior distribution for μ with a mean of 8 minutes and a standard deviation of 0.2 minute. If 10 of these rockets are fired and the first stage has an average burn time of 9 minutes, find a 95% Bayesian interval for μ.

18.8 The daily profit from a juice vending machine placed in an office building is a value of a normal random variable with unknown mean μ and variance σ^2. Of course, the mean will vary somewhat from building to building, and the distributor feels that these average daily profits can best be described by a normal distribution with mean $\mu_0 = \$30.00$ and standard deviation $\sigma_0 = \$1.75$. If one of these juice machines, placed in a certain building, showed an average daily profit of $\bar{x} = \$24.90$ during the first 30 days with a standard deviation of $s = \$2.10$, find

(a) a Bayes estimate of the true average daily profit for this building;

(b) a 95% Bayesian interval of μ for this building;

(c) the probability that the average daily profit from the machine in this building is between \$24.00 and \$26.00.

18.9 The mathematics department of a large university is designing a placement test to be given to incoming freshman classes. Members of the department feel that the average grade for this test will vary from one freshman class to another. This variation of the average class grade is expressed subjectively by a normal distribution with mean $\mu_0 = 72$ and variance $\sigma_0^2 = 5.76$.

(a) What prior probability does the department assign to the actual average grade being somewhere between 71.8 and 73.4 for next year's freshman class?

(b) If the test is tried on a random sample of 100 students from the next incoming freshman class, resulting in an average grade of 70 with a variance of 64, construct a 95% Bayesian interval for μ.

(c) What posterior probability should the department assign to the event of part (a)?

18.10 Suppose that in Example 18.7 the electrical firm does not have enough prior information regarding the population mean length of life to be able to assume a normal distribution for μ. The firm believes, however, that μ is surely between 770 and 830 hours, and it is thought that a more realistic Bayesian approach would be to assume the prior distribution

$$\pi(\mu) = \frac{1}{60}, \quad 770 < \mu < 830.$$

If a random sample of 25 bulbs gives an average life of 780 hours, follow the steps of the proof for Example 18.5 to find the posterior distribution

$$\pi(\mu \mid x_1, x_2, \ldots, x_{25}).$$

18.11 Suppose that the time to failure T of a certain hinge is an exponential random variable with probability density

$$f(t) = \theta e^{-\theta t}, \quad t > 0.$$

From prior experience we are led to believe that θ is a value of an exponential random variable with probability density

$$\pi(\theta) = 2e^{-2\theta}, \quad \theta > 0.$$

If we have a sample of n observations on T, show that the posterior distribution of Θ is a gamma distribution

with parameters

$$\alpha = n + 1 \quad \text{and} \quad \beta = \left(\sum_{i=1}^{n} t_i + 2 \right)^{-1}.$$

18.12 Suppose that a sample consisting of 5, 6, 6, 7, 5, 6, 4, 9, 3, and 6 comes from a Poisson population with mean λ. Assume that the parameter λ follows a gamma distribution with parameters $(3, 2)$. Under the squared-error loss function, find the Bayes estimate of λ.

18.13 A random variable X follows a negative binomial distribution with parameters $k = 5$ and p [i.e., $b^*(x; 5, p)$]. Furthermore, we know that p follows a uniform distribution on the interval $(0, 1)$. Find the Bayes estimate of p under the squared-error loss function.

18.14 A random variable X follows an exponential distribution with mean $1/\beta$. Assume the prior distribution of β is another exponential distribution with mean 2.5. Determine the Bayes estimate of β under the absolute-error loss function.

18.15 A random sample $X_1, \ldots, X_n$ comes from a uniform distribution (see Section 6.1) population $U(0, \theta)$ with unknown θ. The data are given below:

$$0.13, 1.06, 1.65, 1.73, 0.95, 0.56, 2.14, 0.33, 1.22, 0.20,$$
$$1.55, 1.18, 0.71, 0.01, 0.42, 1.03, 0.43, 1.02, 0.83, 0.88$$

Suppose the prior distribution of θ has the density

$$\pi(\theta) = \begin{cases} \frac{1}{\theta^2}, & \theta > 1, \\ 0, & \theta \leq 1. \end{cases}$$

Determine the Bayes estimator under the absolute-error loss function.